CCCA Cell-State Analysis — Rendeiro et al. 2020

Chronic Lymphocytic Leukemia · 10x · 18,657 malignant cells · 7 samples / 4 patients

GO / NO-GO GATES

Gate 1 — Mixing & Continuity

NO-GO

Unintegrated iLISI ≈ 1.0 of 7 (complete per-patient segregation); Leiden clusters 99.3% patient-pure; patient identity 99.5% kNN-predictable. Mixing appears only when Harmony is forced (θ≥3), collapsing patient centroids rather than revealing a shared axis. Per-patient archipelago — no continuous, sample-mixed malignant manifold exists to fit.

Gate 2 — Module Geography

NA

Not evaluated — gate chain stops at the Gate 1 NO-GO (§6).

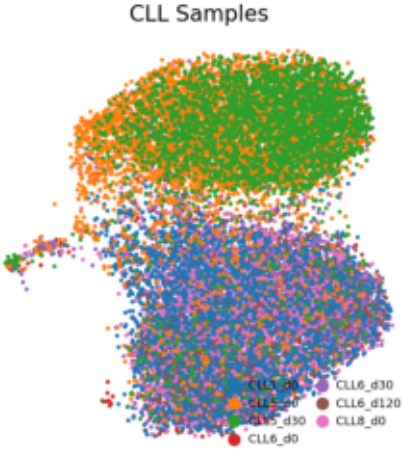
Gate 3 — Hub Cross-Repression

NA

Not evaluated — branching-only; no trajectory fit.

STEP FIGURES

01 · Preprocessing



02 · EMT-TF Expression



Verdict: Pattern does NOT reproduce. The malignant compartment is a set of independent per-patient CLL clones, not a continuous developmental manifold. NO-GO at Gate 1; no trajectory, no bifurcation.

PARAMETER CHOICES & RATIONALE

Sample threshold:	≥100 cells → 7 of 11 samples (18,657 cells) *largest multiplicative count gap 1012→80 (12.6×)*
n_pcs:	20 *scree elbow at 6, floored to atlas robust range (Puram 40 / Kürten 20)*
Integration:	Harmony on sample, θ swept 0→5 *protocol-default θ=2 shown; higher θ over-integrates*
Island removal:	none (inapplicable) *disconnected components are whole patients, not noise*
Geometry:	NA (no trajectory) *manifold not sample-mixed; Gate 1 NO-GO*
EMT-TFs shown:	ZEB2 (12.4%), ZEB1 (3.8%) *only factors >3% detection; SNAI2/TWIST/FOXC2/PRRX2 absent*